Andrea Muolo

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github.com/amuolo

EXPERIENCE

Chief Evangelist - Software Engineer - Business Analyst
Systemorph

Dec 2022 - present Zürich, Switzerland

Project Manager, organizing the platform development to meet client needs
Organize teaching material and conducting workshop with the client
Contributing to the development of the Systemorph Cloud platform and product

Junior Business Analyst
Systemorph

Dec 2021 - Nov 2022 Zürich, Switzerland

Develop the most appropriate data model and formulate the actuarial methodology to strengthen and make business processes efficient

Postdoctoral Research Fellow
Hebrew University of Jerusalem

Oct 2019 - Nov 2021 Jerusalem, Israel

Exact factorization techniques in molecular physics
Genuine predictions on the molecular Berry phase

Research and Teaching Assistant
ETH Zürich

Dec 2018 - Sept 2019 Zürich, Switzerland

Many-body Dirac Equation and relativistic quantum mechanics
Pre-Born–Oppenheimer Tensor Network Algorithms

EDUCATION

Doctor of Sciences
ETH Zürich

Nov 2014 – Dec 2018 Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem

Second level Master's Degree in Science and Technologies
Scuola Universitaria Superiore di Pavia (IUSS)

Oct 2009 – Sept 2014 Pavia, Italy

Focus on transversal and interdisciplinary culture, with a proactive approach
Face the future with responsibility and confidence to build innovative and sustainable projects and solutions

BSc - MSc in Physical Chemistry
University of Pavia and University of Cambridge

Sept 2009 – Sept 2014 Pavia, Italy – Cambridge UK

Th. MSc : A classical dynamic that conserves the quantum Boltzmann distribution
Th. BSc: Molecular Memories: rotaxane- and catenane-based molecular devices

MOST PROUD OF

Swiss National Science Foundation (SNSF) Fellowship
Research Grant, October 2019 - Nov 2021

Project Manager - from Aug 2023
Define scope, plan and organize the execution of the POC.

PUBLICATIONS SELECTION OF

J. Chem. Theory Comput., 18, 234 (2022)
Quantum Proton Effects from Density Matrix Renormalization Group Calculations

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 152, 204103 (2020)
Nuclear-electronic all-particle density matrix renormalization group

J. Chem. Phys., 151, 154110 (2019)
H_3^+ as a five-body problem described with explicitly correlated Gaussian basis sets

J. Chem. Phys., 149, 184105 (2018)
Explicitly correlated Gaussian functions with shifted-center and projection techniques in pre-Born-Oppenheimer calculations

Chimia, 70, 244 (2016)
New Approaches for *ab initio* Calculations of Molecules with Strong Electron Correlation

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: "Matsubara dynamics"

COMPUTER SKILLS

C++, Armadillo, Boost, Eigen	●●●●●
C# 8, ML.NET, Q#	●●●●●
OpenMP 4.0, MPI 3.1	●●●●●
Mathematica, LaTex	●●●●●
Eclipse, Visual Studio, ReSharper, Git	●●●●○
Linux	●●●●○
Python, OpenACC	●●●○○

COMPETENCES

Optimization | Parallel computing | Scientific writing | Programming | Debugging | Code Profiling | Teamwork | Fast learning | Problem solving | Physics | Numerical methods | Teaching | Public speaking/presentation | scrum

LANGUAGES

Italian	(mother tongue)	●●●●●
English	(C1)	●●●●●
German	(A2)	●●○○○

INTERESTS

Sailing, rowing, swimming, dancing
Diving into different points of view
Cooking for dinner with friends
Genuine curiosity and enthusiasm